

AIFS Cross-Attention Mechanism Detail

TimeSeries Tokens ↔ Mistral-7B-Instruct Embeddings Fusion

Mathematical Formulation

Multi-Head Cross-Attention:

1. Projection: $Q = X_{\text{text}} W_Q$, $K = V = X'_{\text{climate}} W_{K,V}$
2. Multi-Head: $Q_h = Q W_h^Q$, $K_h = K W_h^K$, $V_h = V W_h^V$
3. Attention: $A_h = \text{softmax}(Q_h K_h^T / \sqrt{d_k})$
4. Output: $O_h = A_h V_h$
5. Concatenate: $O = \text{Concat}(O_1, \dots, O_{32}) W_O$
6. Residual: $Y = \text{LayerNorm}(X_{\text{text}} + O)$

Attention Matrix
A: [B, 32, 128, 64]
Text pos × Climate pos
Temperature-scaled
 $\tau = 0.1$ (learnable)

AIFS Cross-Attention Specifications:

Input Dimensions:

- Climate: [B, T, 218] → projected to [B, T, 4096]
- Text: [B, seq_len, 4096] (Mistral-7B dimension)

Multi-Head Configuration:

- Heads: 32 (Mistral-7B-Instruct)
- Per-head dimension: 128 ($4096 \div 32$)
- Total parameters: ~67M for attention layers

Attention Mechanism:

- Query: from text embeddings
- Key/Value: from projected climate embeddings
- Temperature scaling: learnable $\tau \in [0.01, 1.0]$
- Dropout: 0.1 during training

Cross-modal
attention matrix

Input Token Representations

AIFS Climate Embeddings
 $X_{\text{climate}} = [B, T, 218]$
 $B=1, \text{time}=2, d_{\text{model}}=218$
From AIFS encoder output

LLM Text Tokens
 $X_{\text{text}} = [B, \text{seq_len}, d_{\text{llm}}]$
 $B=1, \text{seq_len}$ variable, $d_{\text{llm}}=4096$
From climate queries

Climate tokens
projected to
LLM dimension

Dimension Alignment Layer

Climate Projector
 $W_c: 218 \rightarrow d_{\text{llm}}$
Linear(218, 4096) + LayerNorm

Projected Climate
 $X'_{\text{climate}} = [B, T, d_{\text{llm}}]$
Aligned with LLM dim

Multi-Head Cross-Attention Computation

32 attention heads
parallel computation

Query Projection
 $Q = X_{\text{text}} \cdot W_Q$
 $Q: [B, 128, 4096]$

Key Projection
 $K = X'_{\text{climate}} \cdot W_K$
 $K: [B, 64, 4096]$

Value Projection
 $V = X'_{\text{climate}} \cdot W_V$
 $V: [B, 64, 4096]$

Multi-Head Split
32 heads of $d_k=128$
Parallel computation

Attention Computation
 $A_h = \text{softmax}(Q_h K_h^T / \sqrt{d_k})$
Per-head attention weights

Output Fusion & Integration

Head Concatenation
 $\text{Concat}(O_1, \dots, O_{32})$
 $[B, 128, 4096]$

Output Projection
 $W_O: 4096 \rightarrow 4096$
Linear + Dropout

Fused Embeddings
 $Y = [B, 128, 4096]$
Text enhanced with climate context
Ready for Mistral decoder